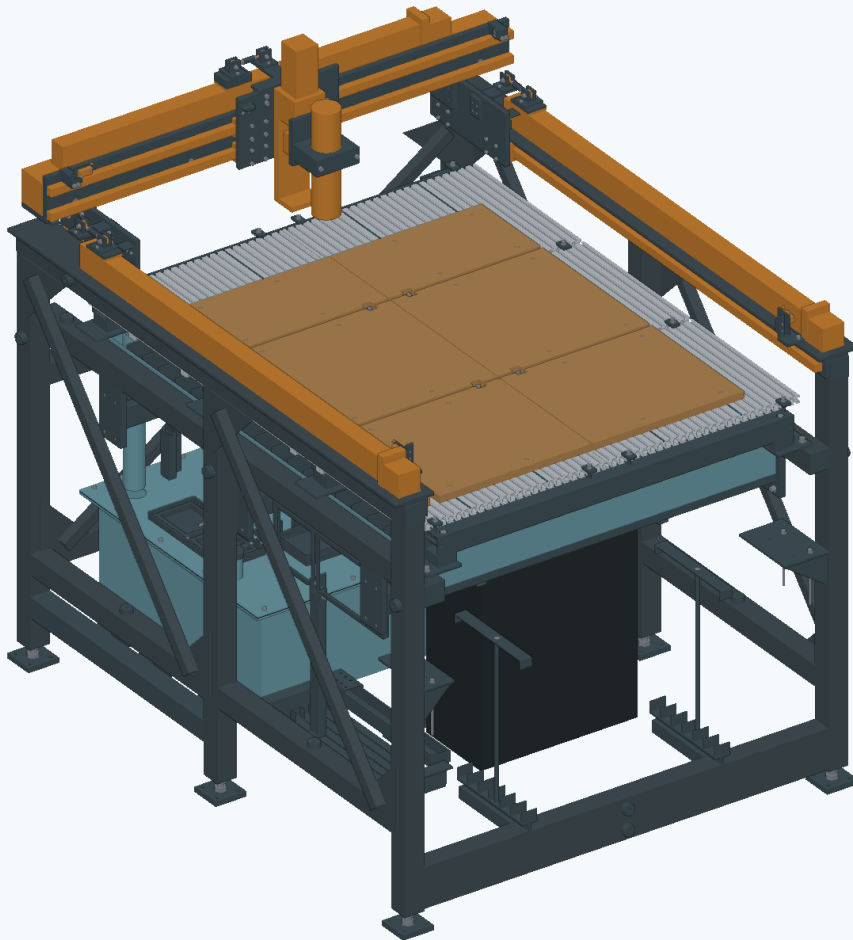


Refining the machine you will build.

Actual Rev H CAD. The Y modules are ordered; the X and Z purchase is planned for Monday. The chassis remains based on 2-inch square tube.



ROUTER BED INSTALLED

Rendered from exported geometry. Amber parts represent purchased-component envelopes; several mating dimensions still require measurement.

800 X / 1000 Y / 100 Z

Nominal travel. Both selected HMS40 X/Y listings specify 10 mm lead: 10 mm movement per screw revolution. This is not screw diameter.

2 x 2-inch steel chassis

Current model wall: 0.120 inch (3.048 mm). Scrap lengths, wall and condition are still unknown. Sand can add stationary mass; it does not supply structural stiffness.

What this revision changes

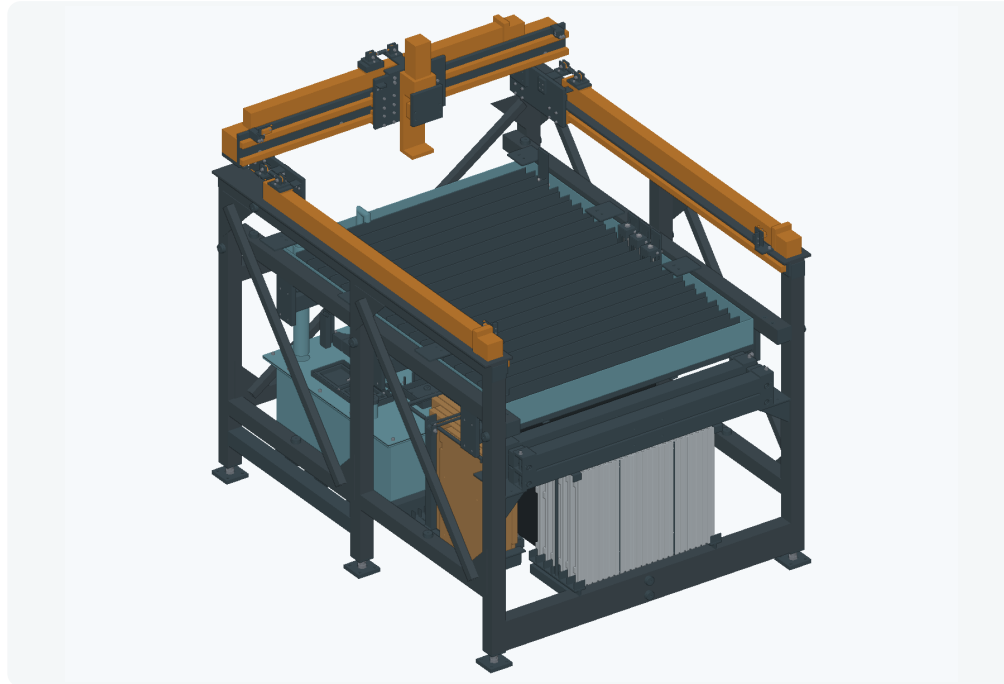
Captive bed nuts, storage restraint details, actual tank service openings and a guarded Z-adapter blank replace several unfinished or misleading details.

Design review, not finished machinery.

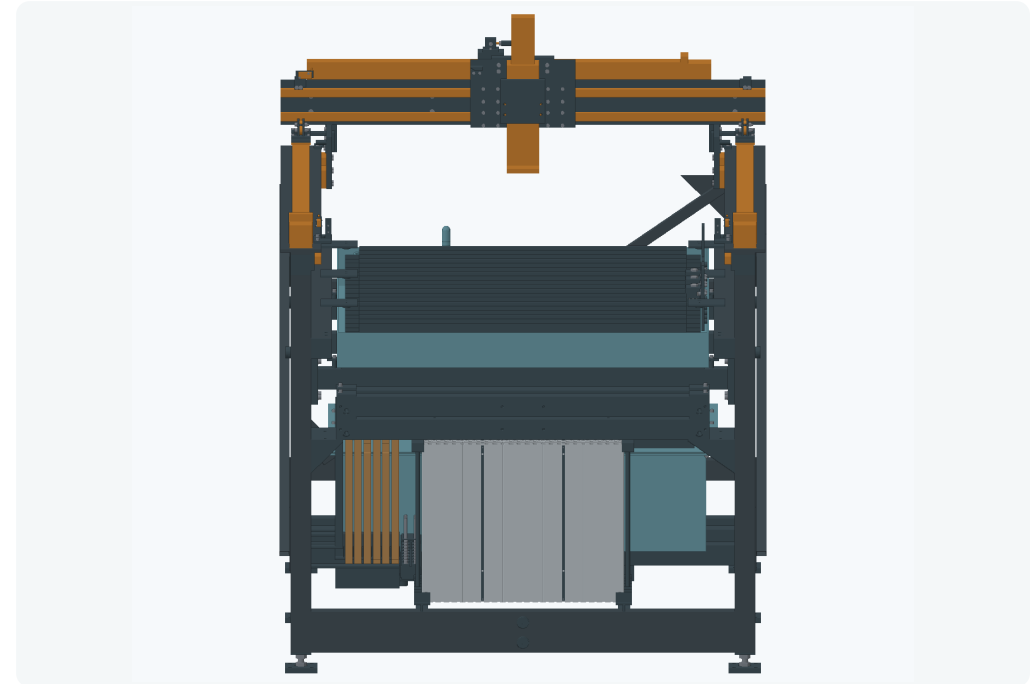
The actual torch head, purchased interfaces and complete machine load verification remain open.

Conversion stays inside the frame.

A manual panel-and-beam bed preserves the required floor footprint. Added retainers address loose parts and storage; conversion still requires multiple fasteners.



STORED BED / ISOMETRIC



STORED BED / FRONT

Scope: the stored view contains the bed and removed router tools. It does not depict a working plasma head. Reinstall bed parts to clear the racks before servicing the controls cabinet.

01 / Captive hardware

Twelve nut strips replace 24 loose spoilboard nuts. Retainers keep strips in the extrusion slots during handling. The exact purchased slot still needs an actual fit check.

02 / Defined storage

Panels, spoilboards and four beams have assigned internal locations. The current design adds restraint hardware; release and staging steps are part of conversion, not optional extras.

03 / Known limitations

Manual handling remains the working assumption. Check access, fingers, tolerances and loaded behavior with the actual parts. The design is not an automatic bed changer.

Buildable details, visible assumptions.

The steel stock lengths remain unchanged. Service openings and fabricated hardware are modeled; unknown supplier interfaces remain explicitly guarded.

BED, STOCK AND WATER

Bare bed	1003 x 1211 overall; larger than nominal cutting travel
Six removable panels	500 x 397 each; separate reachable spoilboards
Extrusion stock	10 bars at 1220 = five two-packs; 3 x 397 per bar
2 x 2 tube blanks	30 blanks / 29,472 mm / 96.69 ft net
Tube lengths and quantities	4 x 1450; 7 x 1048.4; 3 x 1016.4; 6 x 949.2; 4 x 924; 6 x 648.8
Nominal wall	3.048 mm; do not silently substitute thinner scrap
Water storage	Vented reservoir; gravity drain and electric refill
Refill discharge	Air gap above the pan; verify at least 25 mm as built

Net tube length excludes saw loss and other profiles, plate and sheet. Five full 20-ft lengths have a documented nest; random scrap must be nested from its actual usable lengths. No new material price is asserted.

Routine reservoir cleanout

Two gasketed service hatches open the clarified and settling compartments. Covers park in internal pockets. A bolted washout flange replaces the earlier undefined end cap.

Refill path made concrete

A fabricated pipe and stay define the pan entry and discharge. Purchased pump ports, drain valve fittings and hose support still need final supplier dimensions and wet testing.

Adapter drawn as a transfer blank

The 110 x 110 x 12.7 blank retains known custom clamp holes. Unverified Z-carriage slots are removed. Record the actual output pattern before generating those mounting holes.

Power-off Z holding option

The model reserves an optional 57 x 57 x 116.5 braked motor. Its static brake rating is sourced, but coupling, flange fit and stopping behavior are not qualified.

Evidence and the remaining work.

A valid export is useful evidence. It cannot replace actual mounting dimensions, a complete plasma head or a qualified tool-to-work load path.

CHECKED ARTIFACTS

Router state	1448 components; 0 unresolved nominal overlaps
Stored state	1448 components; 0 unresolved nominal overlaps
STEP readback	Both assemblies reimported; matching solid counts and volumes
Source identity	Build manifest records exact source hashes and no changes during generation
Check scope	Geometry, assembly poses and applicable handling/service paths; see the linked completion evidence
Not established	Strength, weld capacity, alignment, cutting accuracy, operator access or electrical qualification

Current package: output/release-review/RevH-CAD
Completion evidence: output/design-completion-2026-09-26
Rebuild: build_revh.py in the legacy-named source directory.

Public project: github.com/DeadlySn777/plasma-router-stand

Y arrival and X/Z order

Measure base slots/nuts, carriage hole pattern, height and stroke datums. The Z listing alone does not resolve its output mounting or prove the optional brake motor will fit.

Actual plasma assembly

The VIV ARC CUT-50 still needs a verified version, start interface and actual torch measurements. Floating touch-off, breakaway and torch-lead clearance remain unfinished.

Rigid structure with the actual stock

Confirm remaining scrap wall and usable lengths. Qualify the complete frame, bed, gantry, mounts, joints and feet against explicit loads and a tool-to-work deflection target.

Operation and procurement

Complete hoses, wiring, enclosure access/thermal design and physical interlock tests. Reconcile current quantities after the interfaces and scrap inventory are known; historical budgets are not a current delivered total.